

List Comprehension

In []: List Comprehension offeres shorter syntax when you want create new **list** based on the values of existing **list**.

Syntax: -newlist=[expression **for** var **in** sequence **if** condition==True]
 newlist=[expression **for** var **in** sequence]

```
In [1]: list1=[4,5,6,8,9]
        for i in list1:
            print(i**3)
```

```
64
125
216
512
729
```

```
In [4]: list1=[4,5,6,8,9]
        cube_list=list() #[]
        for i in list1:
            cube_list.append(i**3)
            print(cube_list)
```

```
[64]
[64, 125]
[64, 125, 216]
[64, 125, 216, 512]
[64, 125, 216, 512, 729]
```

```
In [5]: sub_names=['Hin','Bio','Eng','Math','Che']
        new_list=[]
        for sub in sub_names:
            new_list.append(sub.upper())
            print(new_list)
```

```
['HIN']
['HIN', 'BIO']
['HIN', 'BIO', 'ENG']
['HIN', 'BIO', 'ENG', 'MATH']
['HIN', 'BIO', 'ENG', 'MATH', 'CHE']
```

```
In [7]: list1=[4,5,6,7,8]
        sq=list() #[]
        for i in list1:
            sq.append(i**2)
            print(sq)
```

```
[16]
[16, 25]
[16, 25, 36]
[16, 25, 36, 49]
[16, 25, 36, 49, 64]
```

```
In [6]: list1=[4,5,6,7,8]
square=[i**2 for i in list1]
square
```

```
Out[6]: [16, 25, 36, 49, 64]
```

```
In [8]: list1=[4,5,6,7,8]
square=[i**3 for i in list1]
square
```

```
Out[8]: [64, 125, 216, 343, 512]
```

```
In [9]: sub_names=['Hin','Bio','Eng','Math','Che']
new_list=[sub.upper() for sub in sub_names]
new_list
```

```
Out[9]: ['HIN', 'BIO', 'ENG', 'MATH', 'CHE']
```

```
In [10]: num=[3,2,4,5,6,7,8,9,10,]
even_num=[]
for i in num:
    if i%2==0:
        even_num.append(i)
print(even_num)
```

```
[2, 4, 6, 8, 10]
```

```
In [11]: num=[3,2,4,5,6,7,8,9,10,]
even_list=[i for i in num if i%2==0]
even_list
```

```
Out[11]: [2, 4, 6, 8, 10]
```

```
In [13]: num=[3,2,4,5,6,7,8,9,10,]
odd_list=[i for i in num if i%2!=0]
odd_list
```

```
Out[13]: [3, 5, 7, 9]
```

```
In [14]: days_list=["Monday","Tuesday","Wednesday","Thursday","Friday","Saturday","Sunday"]
result=[]
for day in days_list:
    if len(day)==6:
        result.append(day)
print(result)
```

```
['Monday', 'Friday', 'Sunday']
```

```
In [15]: days_list=["Monday","Tuesday","Wednesday","Thursday","Friday","Saturday","Sunday"]
result=[day for day in days_list if len(day)==6]
result
```

```
Out[15]: ['Monday', 'Friday', 'Sunday']
```

In []: